Uncertainty-based risk ordering and objective-specific ligand ranking for metal–ligand selectivity

Heterogeneous literature measurements

CMI M4 collection: 43,388 records, 4,347 ligands
ligand + metal + oxidation state + medium + temperature

DS1 absolute binding

predict $\log K$, then subtract
two predictions per pair

$\sigma_\Delta = \sqrt{\sigma_X^2 + \sigma_Y^2}$

MoP ACE 0.378

DS2 pairwise selectivity

predict $\Delta\log K$ directly
88,525 rows, 2,503 species pairs
native uncertainty on the target

MoP ACE 0.153

2.5× better calibrated on the quantity the experiment uses

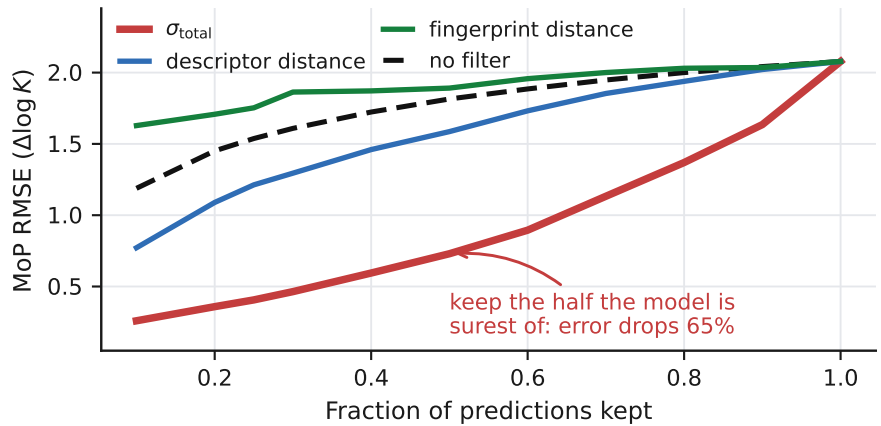
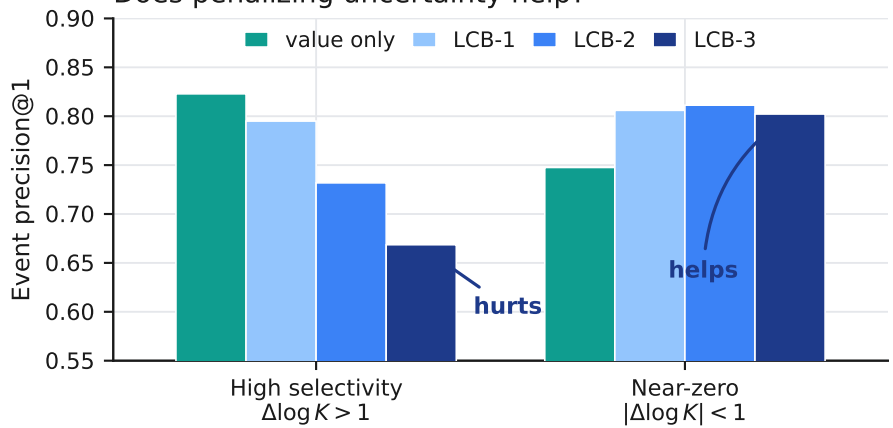
ligand-disjoint:
every test ligand
unseen in training

Fixed direct-UQ workhorse

heteroscedastic MLP ensemble, chosen once
five-fold ligand-disjoint CV, pair-equal (MoP) metrics

Two questions follow ⟶

Which predictions can you trust?



Does penalizing uncertainty help?